Lecture 00: Course Introduction

NT120 – Introduction to AI for Computer Networks and Cybersecurity

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Course page: link.uit.edu.vn/NT120

Outline

1. Welcome
2. What This Course Is About
3. Course Goals and Learning Outcomes
4. The Semester Plan
5. Assessment
6. Tools, Materials and Reading
7. Course Regulations and Expectations
8. Getting Started

What You Will Know by the End of Today

- What NT120 covers, and just as importantly what it does not
- The six course learning outcomes, and which sessions serve each one
- How the fifteen sessions are organised into four blocks
- Exactly how you will be assessed, including the format of both examinations
- Which tools and readings you need, and what to do before Session 1

Keep this deck. Everything about deadlines, weights and examination format is in it, and you should not have to ask twice.

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- Questions about content: bring them to class, or use the course page so that everyone sees the answer
- Questions about your personal situation: e-mail the lecturer of your section directly

Course Identity

Code	NT120
Title (VI)	Nhập môn trí tuệ nhân tạo cho mạng máy tính và an toàn thông tin
Title (EN)	Introduction to Artificial Intelligence for Computer Networks and Cybersecurity
Knowledge block	Major foundation
Credits	2 lecture credits, 0 practice credits, 30 hours of independent study
Prerequisites	None
Prior courses	Introduction to Computer Networks; Introduction to Programming
Owning unit	Faculty of Computer Networks and Communications
Sessions	15 sessions, 90 minutes each, all activities included

There is no separate laboratory credit. Guided practice happens inside the 90 minutes and in the homework notebooks, and it is assessed as part of component A1.

The Course Info



`link.uit.edu.vn/NT120`

Everything for NT120 lives at Course-UIT

- Slides for every session, posted before the class
- The course overview document: the full session-by-session plan
- Notebooks and datasets for the guided practice
- Homework instructions, deadlines and the submission link
- The reading list, with the exact chapters and pages
- Announcements, including anything about the examinations

Scan the code, or type `link.uit.edu.vn/NT120`

Check the page before each session. If something is unclear there, ask on the page rather than by e-mail, so that the whole class sees the answer.

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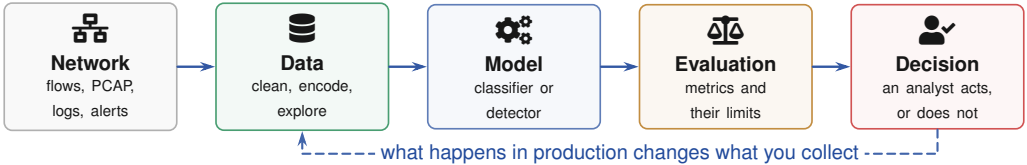
Course Description

NT120 gives you an introductory, working knowledge of artificial intelligence and machine learning, focused on how to

- **define a problem** in a networking or security setting,
- **prepare data** — network traffic, logs and security events,
- **build basic models** for classification and anomaly detection, and
- **evaluate and interpret** the results honestly.

You practise on small, guided datasets of flow records, logs and security events. Logistic regression and decision trees are treated properly; k-NN, naive Bayes and support vector machines are introduced briefly so that you can recognise where each applies. Deep learning, Transformers and LLMs, federated learning, reinforcement learning and deep RL are presented at *orientation level*, linked to the advanced courses that develop them. The course also addresses adversarial AI, privacy, bias and responsible use.

The Course in One Picture



Sessions 1–2 frame the problem, Sessions 3–5 build the data skills, Sessions 6–11 build and judge the models, Sessions 12–15 open the frontier and the risks.

What This Course Is and Is Not

✔ It is

- Framing a network or security question as a learning task
- Working with flow records, logs and security events as data
- Training baseline models and reading their output honestly
- Recognising the risks: adversarial AI, privacy, bias
- Knowing when the answer is “do not use ML here”

✘ It is not

- A deep learning course: no architectures to implement
- An advanced network security course: no exploitation
- A mathematics course: intuition first, derivations later
- A tour of the newest models for their own sake

If you leave able to *frame* a problem and *distrust* a result, the course has done its job.

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Four Course Goals

Goal	After this course you can . . .	Programme LO
G1	Explain the concepts, the role and the basic workflow of AI/ML in computer networks and cybersecurity	2.1; 2.2.4
G2	Prepare and explore data, and apply basic AI/ML models under guidance	2.2.4; 3.1
G3	Evaluate and interpret results, and select an appropriate method for a simple networking or security problem	3.3; 4.2
G4	Identify the limitations, security risks, privacy and ethical issues of AI, and the directions it is developing in	3.3; 8.1

The right-hand column links each goal to the learning outcomes of the degree programme. You do not need to memorise it; it exists so that the course fits into the curriculum.

Six Course Learning Outcomes

CLO	Description	Level
G1.1	Explain AI/ML concepts, supervised and unsupervised learning, and representative applications	Teach
G1.2	Describe the workflow from problem definition and data preparation through to model evaluation	Teach
G2.1	Preprocess and explore a small networking or security dataset using a guided notebook	Utilise
G2.2	Train a basic classification or anomaly detection model with available tooling	Utilise
G3.1	Interpret the confusion matrix, accuracy, precision, recall and F1-score, and the limits of the result	Utilise
G4.1	Analyse basic risks of adversarial AI, generative AI, privacy and bias, use AI responsibly, and identify advanced study paths	Teach

Level: Introduce $\rightarrow$ **Teach** $\rightarrow$ **Utilise**. “Utilise” means you will be asked to *do* it, not only to describe it.

Which Outcome Is Assessed Where

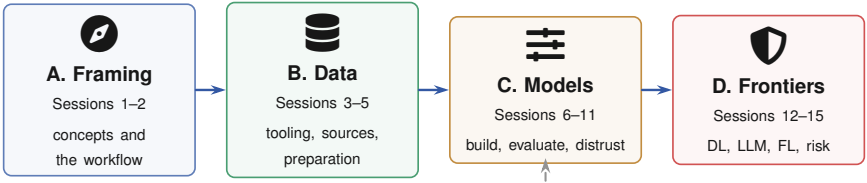
	A1 continuous	A2 midterm	A4 final
G1.1	✓	✓	
G1.2	✓	✓	
G2.1	✓		
G2.2	✓		✓
G3.1		✓	✓
G4.1			✓

A3 (practical) carries 0% on its own: the notebooks are graded inside A1. Every outcome is assessed at least twice.

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Four Blocks



Blocks A to C are cumulative: each depends on the one before it. Block D is orientation, designed to let you read the literature and choose an advanced course, not to make you build those systems.

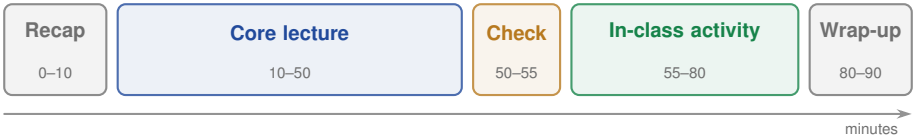
Sessions 1 to 8

S.	Topic	CLOs	Assessed in
1	Overview of AI/ML in networks and security: concepts, method families, typical applications	G1.1	A1
2	Problem definition and the AI/ML workflow: from requirement to data and metric; train/validation/test; human oversight	G1.2	A1
3	Python and Google Colab with tabular data: Pandas, NumPy, reproducible notebooks	G2.1	A1
4	Networking and security data: flows, packets, PCAP, logs and events; features, labels, data quality	G1.2, G2.1	A1
5	Preprocessing and exploratory analysis: missing values, encoding, scaling, statistics, visualisation, outliers	G2.1	A1
6	Baseline supervised models: logistic regression and decision trees; brief k-NN, naive Bayes and SVM	G1.1, G2.2	A1
7	Evaluating classifiers: confusion matrix, accuracy, precision, recall, F1; choosing a metric from error costs	G3.1	A1, A2
8	Reliability of results: overfitting, data leakage, class imbalance; the limits of data and model	G3.1	A1

Sessions 9 to 15

S.	Topic	CLOs	Assessed in
9	**Midterm examination** and feedback: foundations, data, model evaluation and reliability; common mistakes	G1.1, G1.2, G3.1	A2
10	Introduction to anomaly detection: classification versus anomaly detection; Isolation Forest and its uses	G1.1, G2.2	A1
11	**Integrated mini-lab**: preprocessing, baseline model, evaluation and interpretation on an intrusion or anomaly problem	G2.1, G2.2, G3.1	A1
12	Deep learning and Transformers overview: MLP, CNN/RNN, autoencoders; what Transformers solve; links to advanced courses	G4.1	A4
13	LLMs and federated learning: log analysis and threat intelligence; hallucination, prompt injection and data leakage; privacy-preserving distributed learning	G4.1	A4
14	Adversarial AI, RL/DRL and responsible AI: evasion, poisoning and defences; RL for network optimisation; privacy, bias and ethics	G4.1	A4
15	Synthesis, study pathways and revision: method selection, evaluation and limits; links to later courses	G2.2, G3.1, G4.1	A4

What a Typical 90 Minutes Looks Like



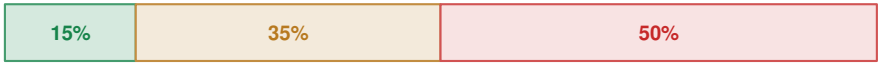
Recap answers to the previous homework and one motivating question from practice • **Core lecture** new concepts, worked examples, live notebook • **Check** two or three questions answered by the whole class • **Activity** guided pair or small-group work • **Wrap-up** takeaways and the homework briefing

Sessions 9 and 11 break this pattern: Session 9 is the midterm examination plus feedback, Session 11 is the integrated mini-lab.

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How the Mark Is Made Up



A1 — Continuous

quizzes, exercises, notebooks, class discussion

G1.1, G1.2, G2.1, G2.2

A2 — Midterm

Session 9, on paper, closed book, multiple choice

G1.1, G1.2, G3.1

A4 — Final

examination period, on paper, closed book, multiple choice and essay

G2.2, G3.1, G4.1

A3 — Practical: 0%. The course has no separate practice credit. Your notebooks and in-class work are graded inside A1, against the same rubric.

The Two Examinations

A2 — Midterm, Session 9

- **On paper**, in the classroom
- **Closed book**: no notes, no textbook, no device
- **Multiple choice** only
- Scope: Sessions 1–8
- **35%** of the course mark
- Marked answers are discussed in the same session

A4 — Final examination

- **On paper**, in the examination period
- **Closed book**: no notes, no textbook, no device
- **Multiple choice *and* essay**
- Scope: the whole syllabus, Sessions 1–15
- **50%** of the course mark
- Format and question styles are previewed in Session 15

Both examinations are handwritten on paper with the book closed. Plan your revision accordingly: you must be able to compute a confusion matrix and its metrics without a laptop.

What the Essay Questions Look For

The final examination's essay section is not a memory test. It asks you to choose and defend.

- **Method selection.** Given a described network and threat, choose a paradigm, a model family and a metric, and justify each choice.
- **Analysis of results and limits.** Read a given result, say what it does and does not show, and name the population it generalises to.
- **Risk and responsible use.** Identify the adversarial, privacy and bias risks of a described deployment, and propose proportionate mitigations.

A well-defended imperfect answer scores above an undefended perfect one. Practise writing the *reason*, not only the conclusion.

How A1 Is Marked

Criterion	Excellent (4)	Satisfactory (2)	Poor (0–1)
Understanding of concepts and the workflow	Accurate explanation, correctly connected to a networking or security problem	Main ideas understood but incomplete	Very limited, or absent
Data and model practice	Notebook completed correctly, results reasonable, reasoning explained	Partly completed, needed support	Wrong or not submitted
Exercises, discussion and independent study	Complete, active participation, clear argument	Minimum level of completion	Little participation, or nothing submitted

The full rubrics for A1, A2 and A4 are in the syllabus. The pattern is the same in all three: explaining *why* carries as much weight as producing the number.

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Tools You Will Use

Python 3.10+
the only language used in this course

Google Colab
nothing to install; a browser is enough

Pandas, NumPy
Matplotlib and Seaborn for figures

Scikit-learn
models, pipelines and metrics

- **Wireshark** / **tshark** appear in Session 4 as an instructor demonstration of PCAP data. You are not required to install them.
- **LLM tools and deep learning frameworks** appear in Sessions 12–13 as demonstrations only.

Everything assessed in this course runs in Google Colab. If your laptop cannot install anything, you can still complete every notebook.

Books and References

Principal texts

[1] Gupta, B. B. & Sheng, M. (2019). *Machine Learning for Computer and Cyber Security: Principles, Algorithms, and Practices.* CRC Press.

[2] Géron, A. (2023). *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow* (3rd ed.). O'Reilly.

[3] Chio, C. & Freeman, D. (2018). *Machine Learning and Security: Protecting Systems with Data and Algorithms.* O'Reilly.

Further references

[4] Goodfellow, Bengio & Courville (2016). *Deep Learning.* MIT Press.

[5] Stamp, M. (2022). *Introduction to Machine Learning with Applications in Information Security* (2nd ed.).

[6] Sumeet, D. & Shrey, J. (2024). *Hands-On Artificial Intelligence for Cybersecurity.* Packt.

[7–9] MITRE ATT&CK; MITRE ATLAS; NIST AI Risk Management Framework 1.0 (NIST AI 100-1).

Each session's slides name the exact chapters and pages to read. You are never asked to read a whole book.

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The Rules

Attendance

You must attend at least **70%** of sessions to be eligible for the final examination.

Deadlines

Work submitted after the deadline is **not marked**. Submit what you have rather than nothing.

Examination conduct

Violations of examination regulations are handled under university rules.

- Group work in the mini-lab is submitted as a group, but each member also writes an individual reflection.
- If illness or a personal emergency affects a deadline, contact your lecturer *before* it passes, not afterwards.

Using AI Assistants in This Course

This is a course about AI. You are not forbidden from using it. You are expected to use it the way a professional does.

- **Allowed:** explaining an error message, suggesting a Pandas idiom, checking your reasoning, drafting a comment.
- **Required:** you must be able to explain every line you submit, in class, without help. If you cannot, do not submit it.
- **Declared:** write one line in the notebook saying where you used an assistant and for what.
- **Never:** for the two examinations, which are on paper and closed book.

⚠ Session 13 shows you exactly how confidently an LLM produces wrong security claims. Treat its output as a draft to verify, never as an answer to copy.

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How to Do Well in NT120

Habits that work

- Do the homework notebook the same week; the sessions are cumulative
- Write down *why* you dropped a column, not only that you did
- Ask about anything in a reading you did not follow, at the start of the next session
- Practise the metric calculations by hand: they appear on paper in both examinations

Habits that fail

- Copying a notebook that runs but that you cannot explain
- Reporting a high accuracy without checking the class balance
- Leaving the reading until the week of the midterm
- Treating the mini-lab as a coding exercise rather than an argument about a result

The single best predictor of a good mark in this course is whether you can explain a result you did not expect.

Before Session 1

1. **Open Google Colab** with your university account and run one cell, so that you know it works.
2. **Open the course page** and check that you can see the slides, the notebooks and the datasets.
3. **Skim the course overview document** so that you know where the semester is going.
4. **Bring one question:** a networking or security problem you would like to know whether AI can help with. We use these in the Session 1 activity.



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No technical preparation is required. Session 1 assumes only the two prior courses: Introduction to Computer Networks and Introduction to Programming.

Summary

- NT120 is a 2-credit, 15-session introduction to AI and ML *for networking and security*, built around framing problems, handling data, building baselines and judging results honestly.
- Six learning outcomes, four blocks: framing, data, models, frontiers.
- Assessment: **A1 continuous 15%**, **A2 midterm 35%**, **A4 final 50%**; practical work is graded inside A1.
- Both examinations are **on paper and closed book**: the midterm is multiple choice, the final is multiple choice and essay.
- Everything runs in Google Colab; each session names the exact pages to read.
- Attendance minimum 70%; late work is not marked.

Next session: what artificial intelligence and machine learning actually are, and where they already sit inside network operations and cyber defence.

Questions

About the content

ask in class, or on the course page

About your situation

e-mail your section's lecturer

Welcome to NT120.

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